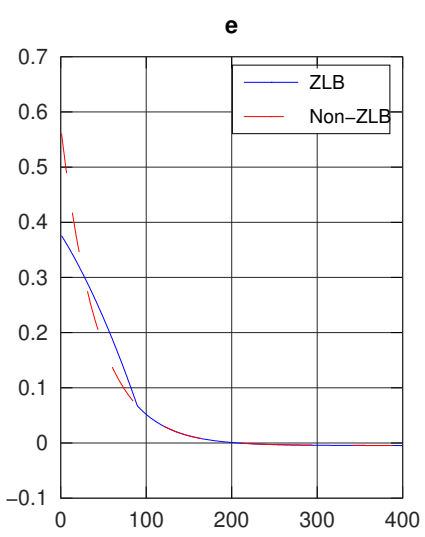
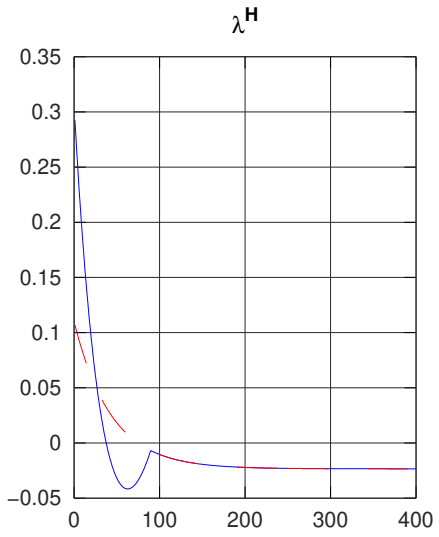
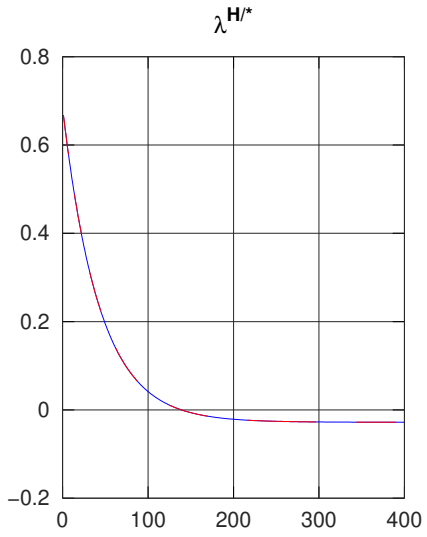
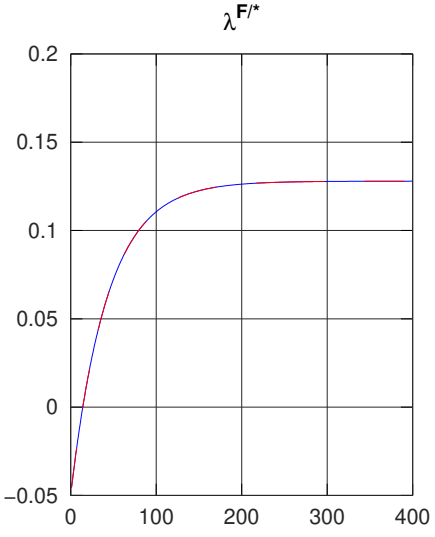
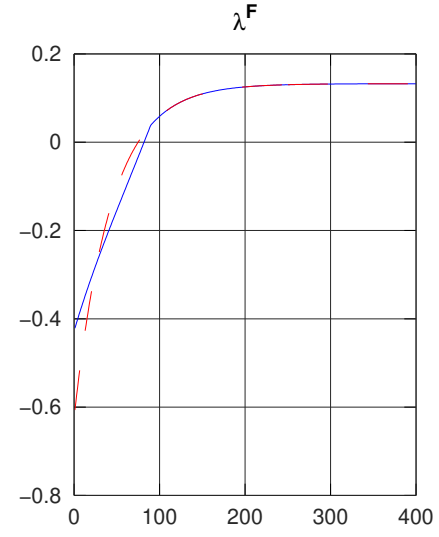
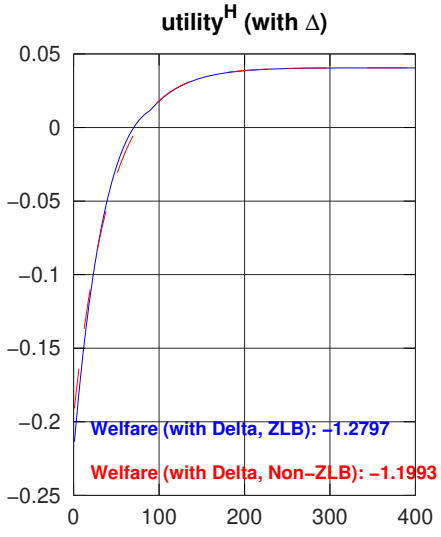

e
ZLB
Non−ZLB
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
−0.1
0
100
200
300
400

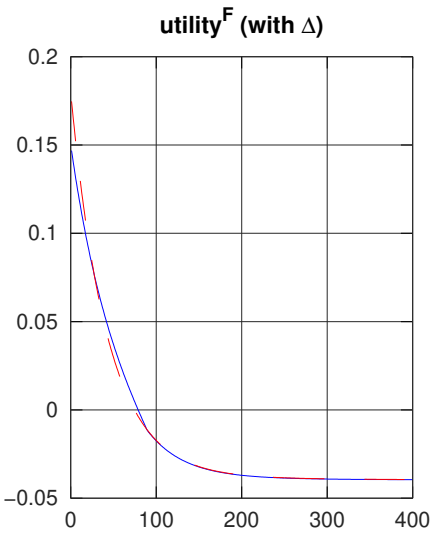

λ^H
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
−0.05
0
100
200
300
400

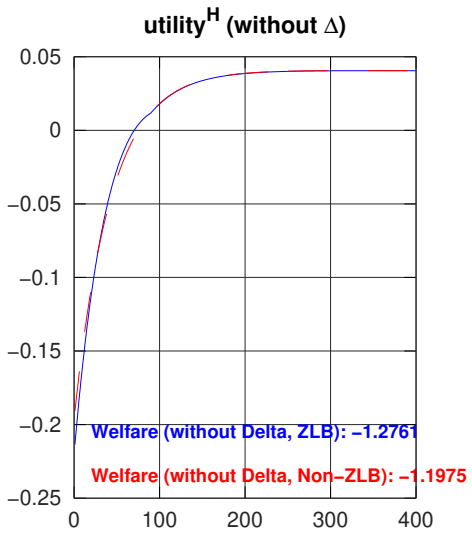

λ^H/*
0.8
0.6
0.4
0.2
0
−0.2
0
100
200
300
400

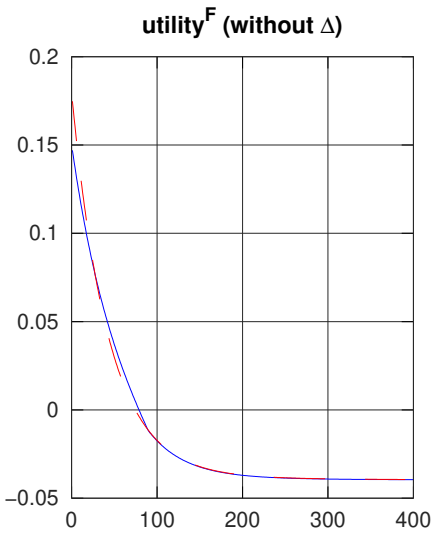

λ^F/*
0.2
0.15
0.1
0.05
0
−0.05
0
100
200
300
400

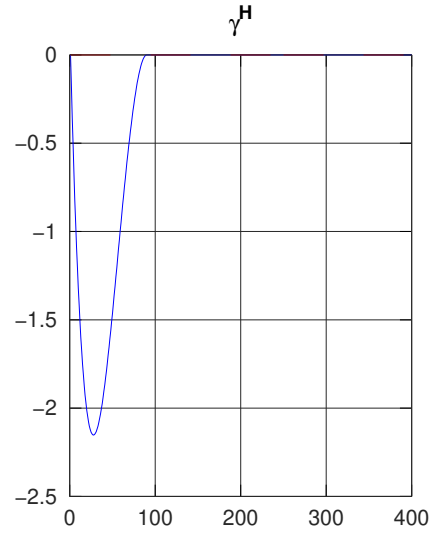

λ^F
0.2
0
−0.2
−0.4
−0.6
−0.8
0
100
200
300
400

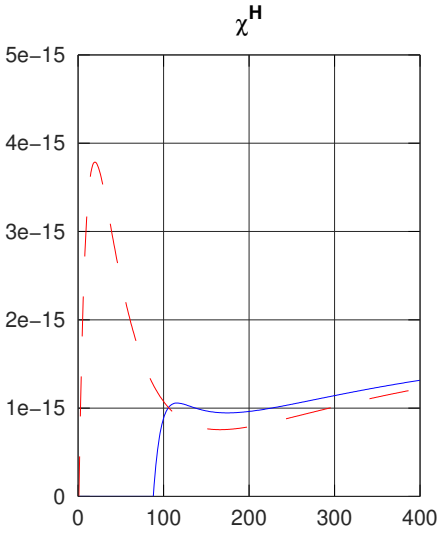

utility^H (with Δ)
0.05
0
−0.05
−0.1
−0.15
−0.2
−0.25
0
100
200
300
400
Welfare (with Delta, ZLB): −1.2797
Welfare (with Delta, Non−ZLB): −1.1993

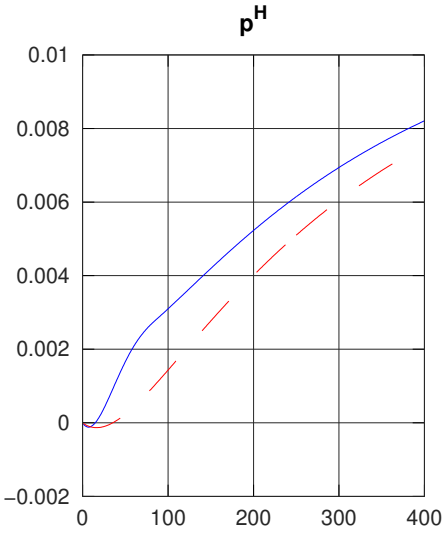

utility^F (with Δ)
0.2
0.15
0.1
0.05
0
−0.05
0
100
200
300
400

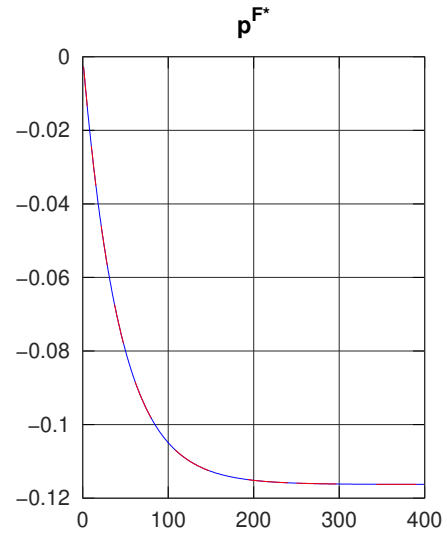

utility^H (without Δ)
0.05
0
−0.05
−0.1
−0.15
−0.2
−0.25
Welfare (without Delta, ZLB): −1.2761
Welfare (without Delta, Non−ZLB): −1.1975
0
100
200
300
400

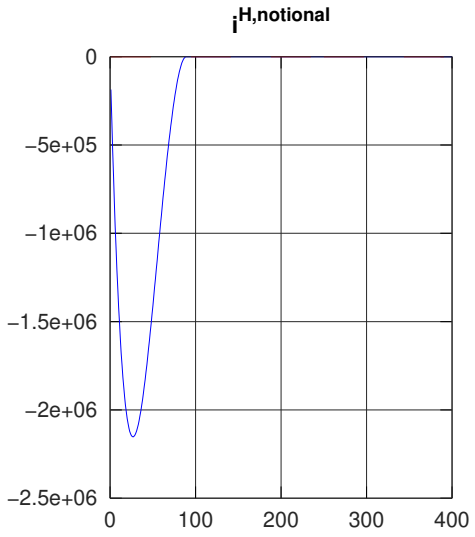

utility^F (without Δ)
0.2
0.15
0.1
0.05
0
−0.05
0
100
200
300
400

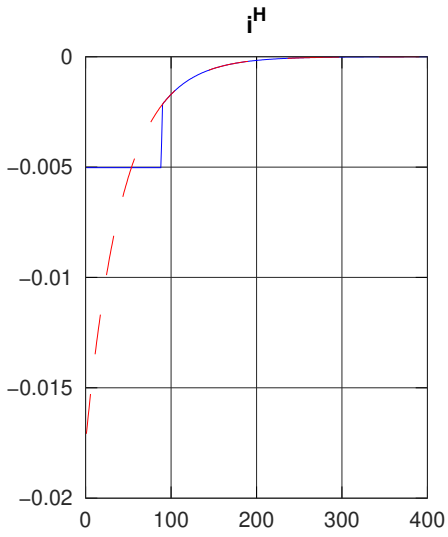

γ^H
0
−0.5
−1
−1.5
−2
−2.5
0
100
200
300
400

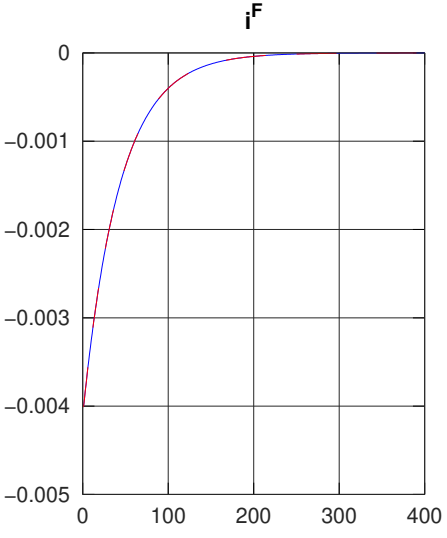

χ^H
5e−15
4e−15
3e−15
2e−15
1e−15
0
0
100
200
300
400

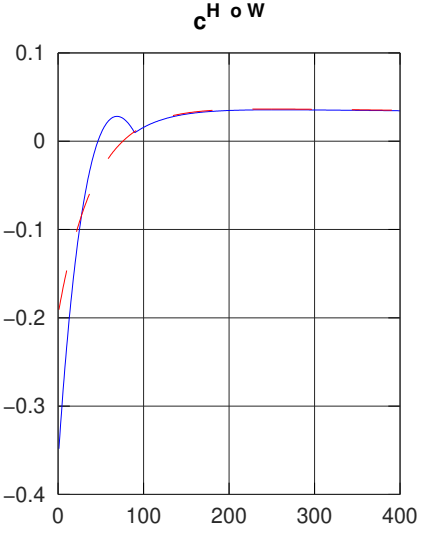

p^H
0.01
0.008
0.006
0.004
0.002
0
−0.002
0
100
200
300
400

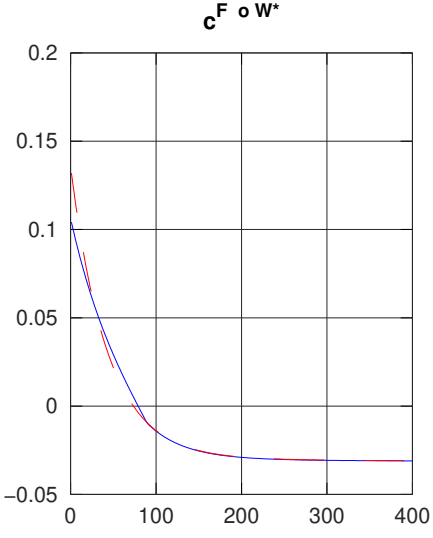

p^{F*}
0
−0.02
−0.04
−0.06
−0.08
−0.1
−0.12
0
100
200
300
400

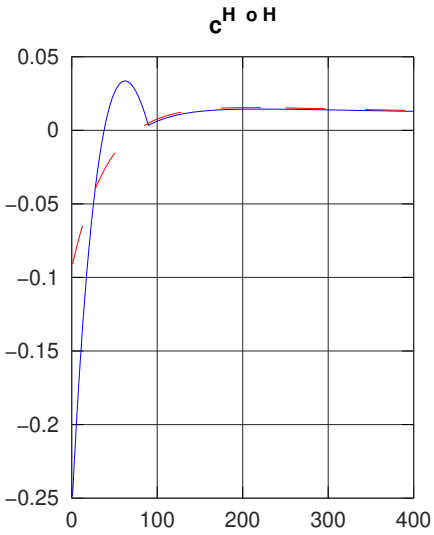

$i^{H,notional}$
0
−5e+05
−1e+06
−1.5e+06
−2e+06
−2.5e+06
0
100
200
300
400

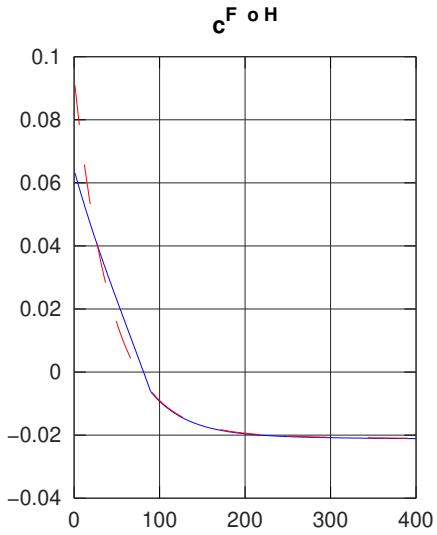

i^{H}
0
−0.005
−0.01
−0.015
−0.02
0
100
200
300
400

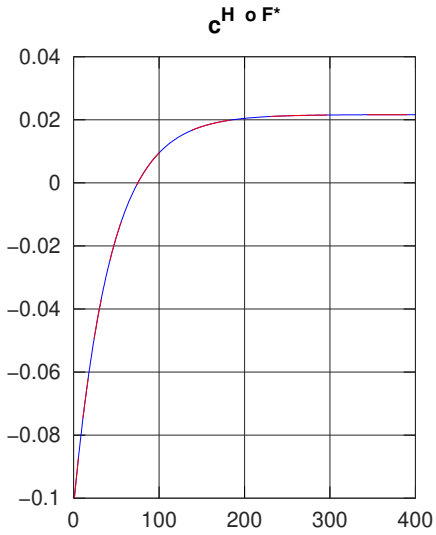

i^{F}
0
−0.001
−0.002
−0.003
−0.004
−0.005
0
100
200
300
400

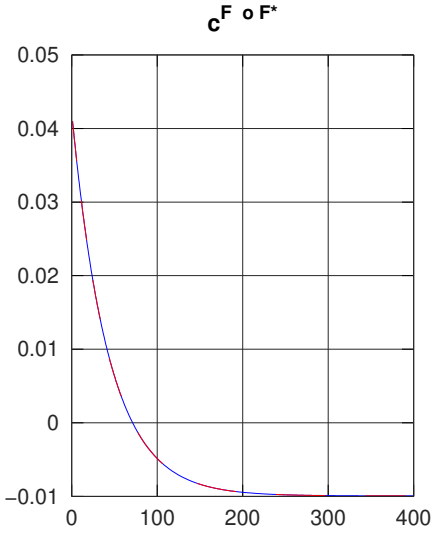

$c^{H\ o\ W}$
0.1
0
−0.1
−0.2
−0.3
−0.4
0
100
200
300
400

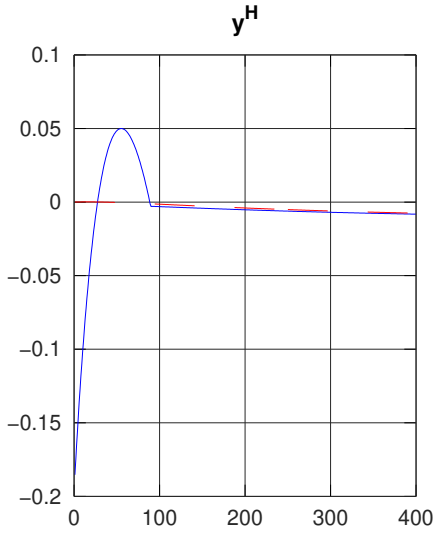

$c^{F\ o\ W*}$
0.2
0.15
0.1
0.05
0
−0.05
0
100
200
300
400

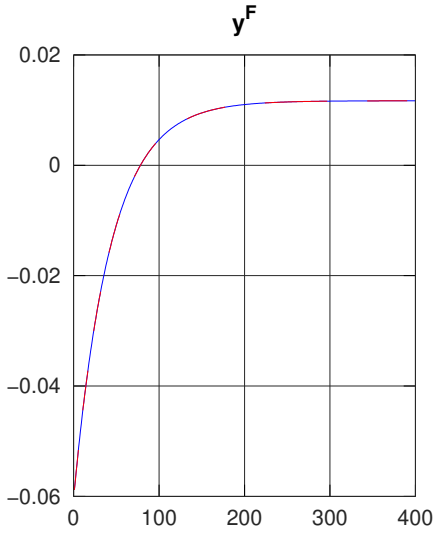

$c^{H \circ H}$
0.05
0
−0.05
−0.1
−0.15
−0.2
−0.25
0
100
200
300
400

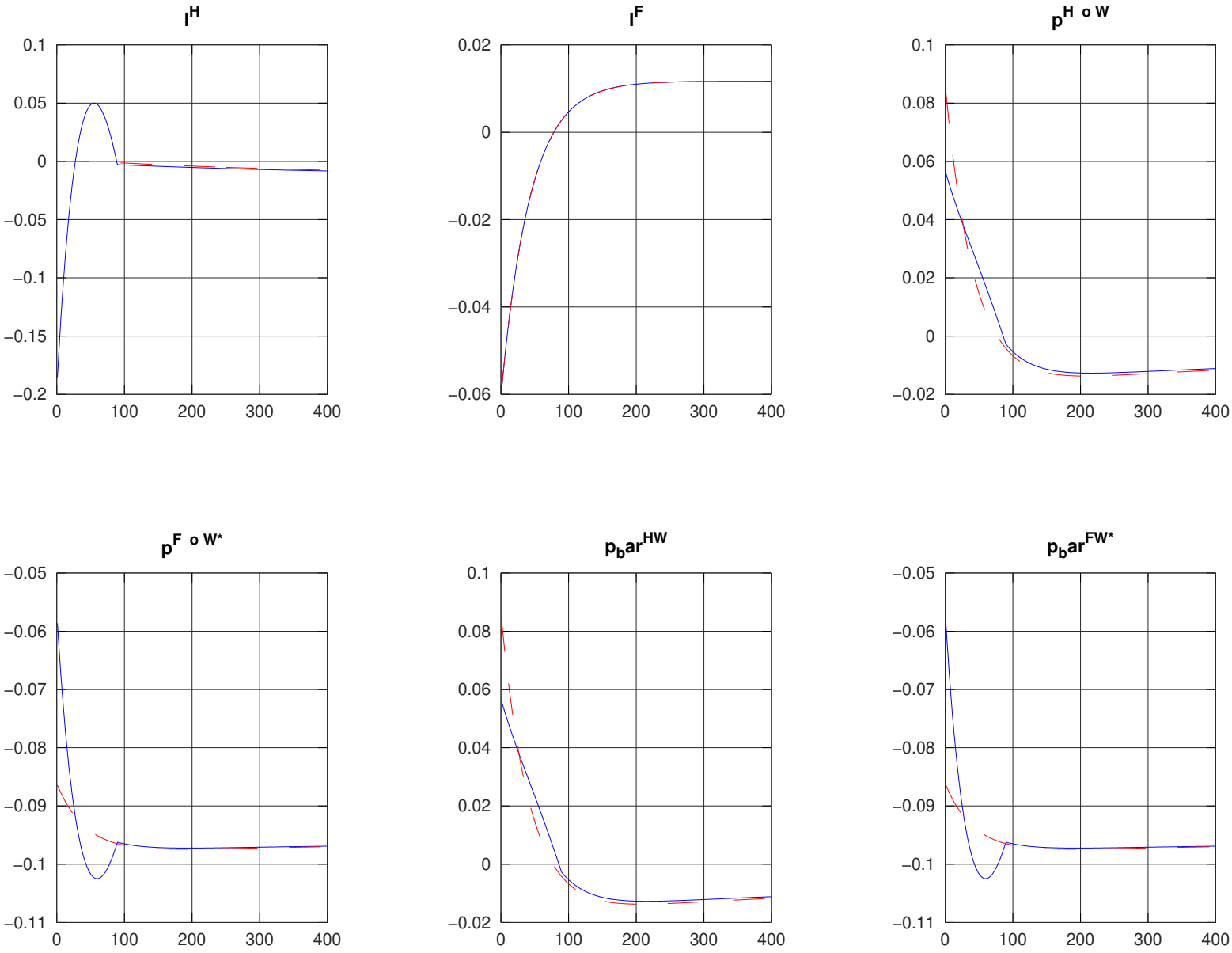

$c^{F \circ H}$
0.1
0.08
0.06
0.04
0.02
0
−0.02
−0.04
0
100
200
300
400

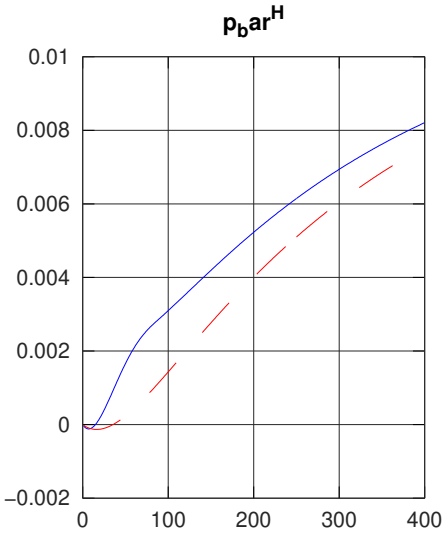

$c^{H \circ F^*}$
0.04
0.02
0
−0.02
−0.04
−0.06
−0.08
−0.1
0
100
200
300
400

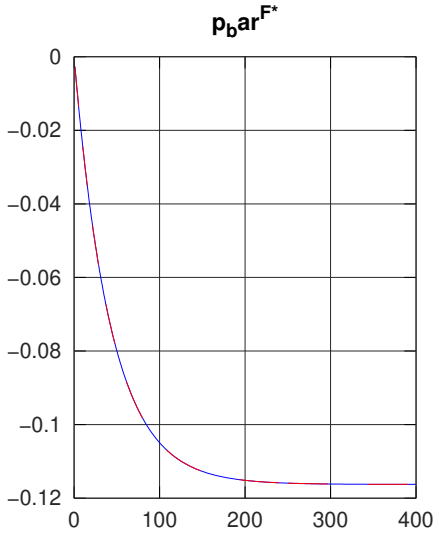

$c^{F \circ F^*}$
0.05
0.04
0.03
0.02
0.01
0
−0.01
0
100
200
300
400

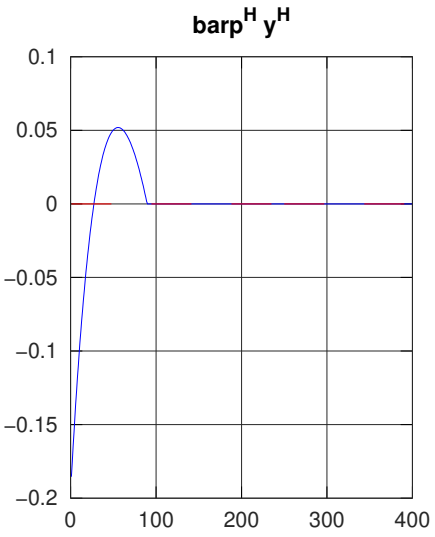

y^H
0.1
0.05
0
−0.05
−0.1
−0.15
−0.2
0
100
200
300
400

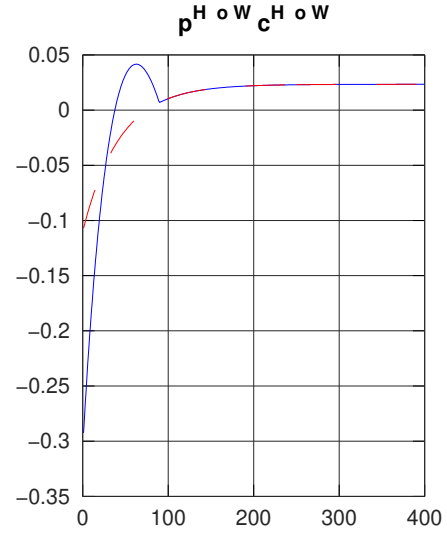

y^F
0.02
0
−0.02
−0.04
−0.06
0
100
200
300
400

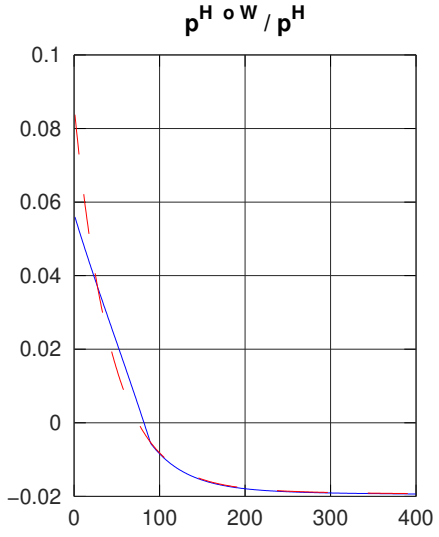

l^H
0.1
0.05
0
−0.05
−0.1
−0.15
−0.2
0
100
200
300
400
l^F
0.02
0
−0.02
−0.04
−0.06
0
100
200
300
400
$p^{H\ o\ W}$
0.1
0.08
0.06
0.04
0.02
0
−0.02
0
100
200
300
400
$p^{F\ o\ W^*}$
−0.05
−0.06
−0.07
−0.08
−0.09
−0.1
−0.11
0
100
200
300
400
p_bar^{HW}
0.1
0.08
0.06
0.04
0.02
0
−0.02
0
100
200
300
400
$p_bar^{FW^*}$
−0.05
−0.06
−0.07
−0.08
−0.09
−0.1
−0.11
0
100
200
300
400

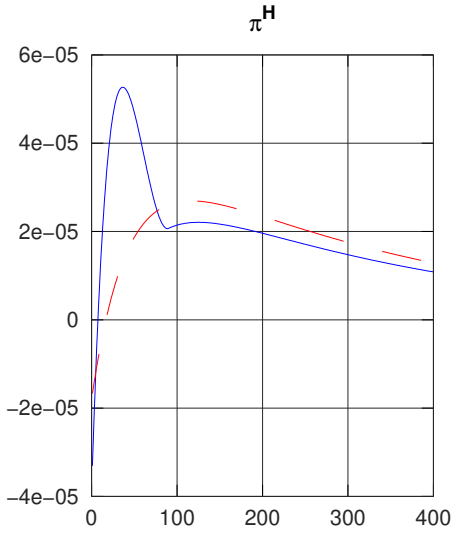

$p_b ar^H$
0.01
0.008
0.006
0.004
0.002
0
−0.002
0
100
200
300
400

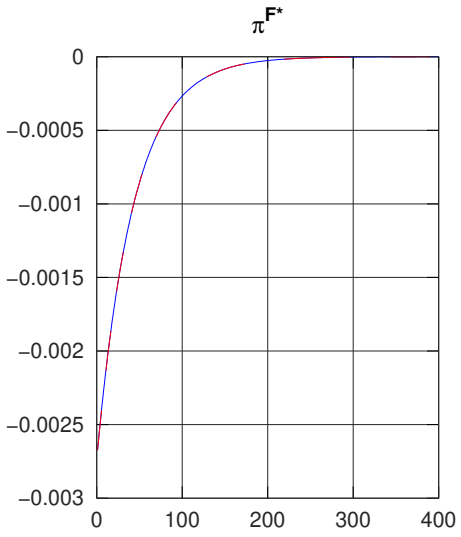

$p_b ar^{F*}$
0
−0.02
−0.04
−0.06
−0.08
−0.1
−0.12
0
100
200
300
400

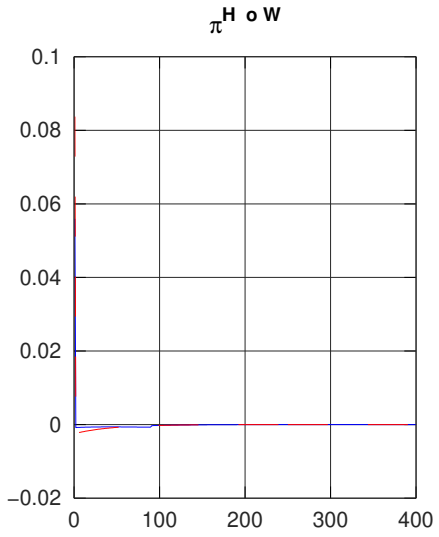

$barp^H y^H$
0.1
0.05
0
−0.05
−0.1
−0.15
−0.2
0
100
200
300
400

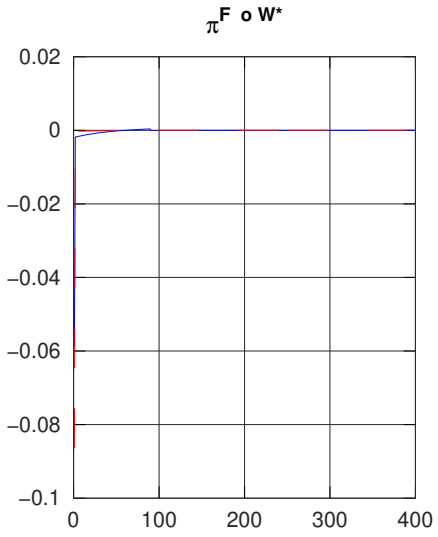

$p^{H\ o\ W} c^{H\ o\ W}$
0.05
0
−0.05
−0.1
−0.15
−0.2
−0.25
−0.3
−0.35
0
100
200
300
400

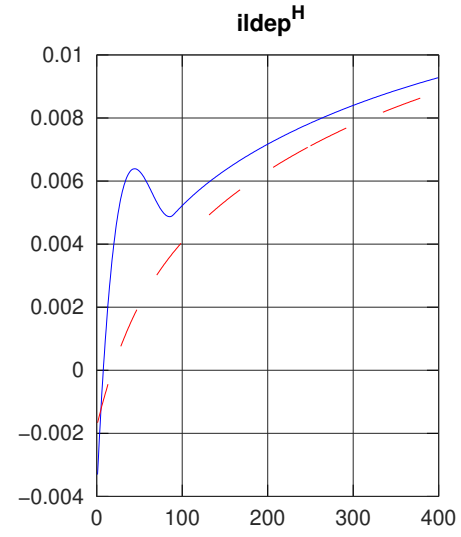

$p^{H\ o\ W} / p^H$
0.1
0.08
0.06
0.04
0.02
0
−0.02
0
100
200
300
400

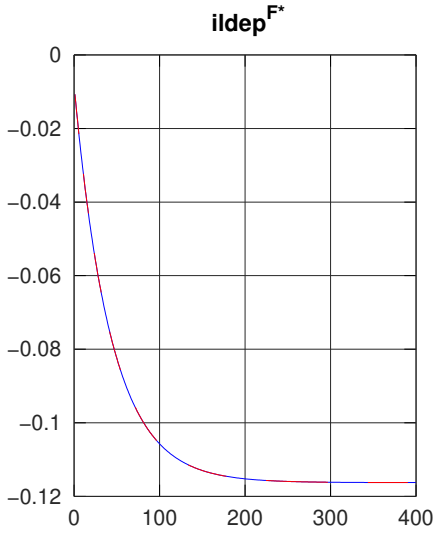

π^H
6e−05
4e−05
2e−05
0
−2e−05
−4e−05
0
100
200
300
400

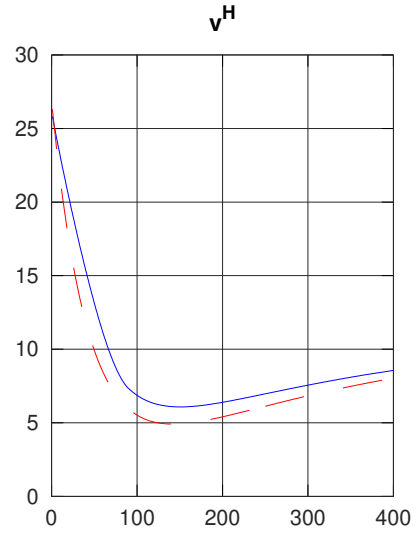

π^{F*}
0
−0.0005
−0.001
−0.0015
−0.002
−0.0025
−0.003
0
100
200
300
400

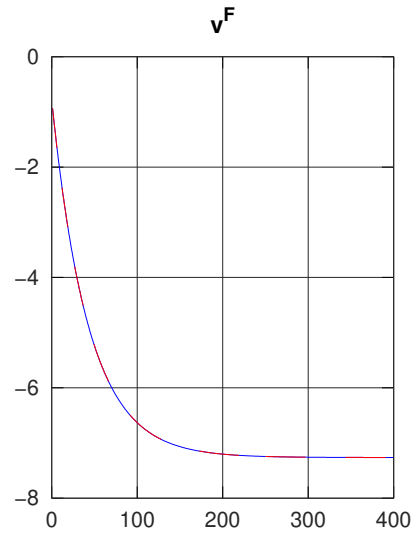

$\pi^{H\ o\ W}$
0.1
0.08
0.06
0.04
0.02
0
−0.02
0
100
200
300
400

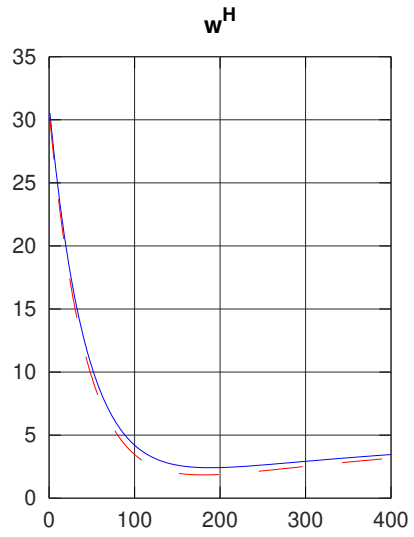

$\pi^{F\ o\ W*}$
0.02
0
−0.02
−0.04
−0.06
−0.08
−0.1
0
100
200
300
400

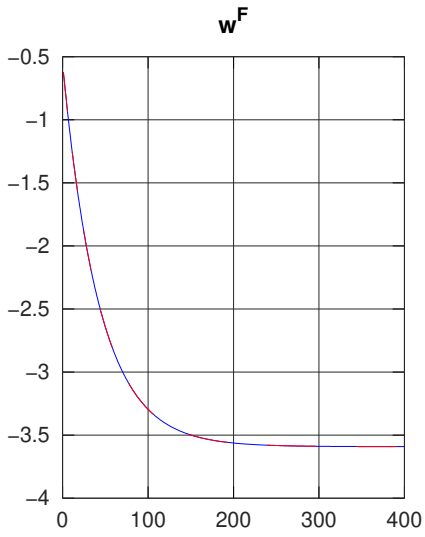

$ildep^{H}$
0.01
0.008
0.006
0.004
0.002
0
−0.002
−0.004
0
100
200
300
400

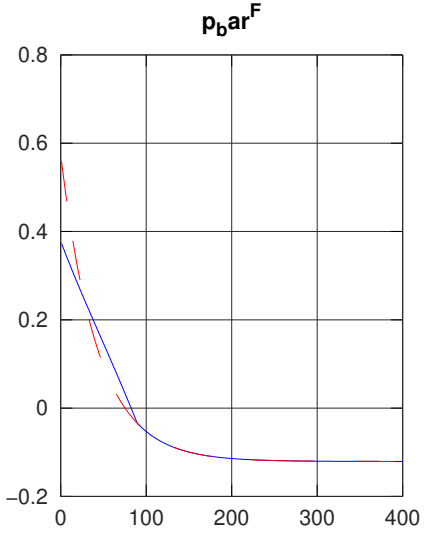

$ildep^{F*}$
0
−0.02
−0.04
−0.06
−0.08
−0.1
−0.12
0
100
200
300
400

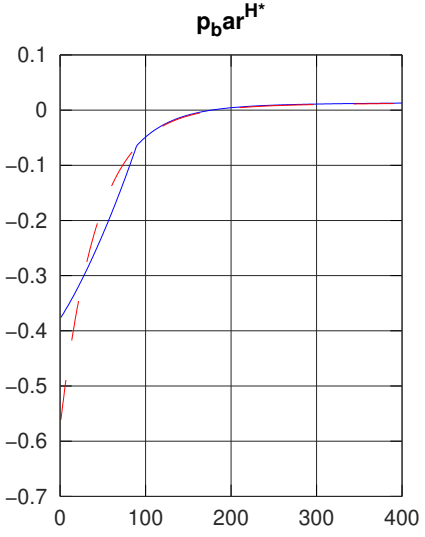

v^{H}
30
25
20
15
10
5
0
0
100
200
300
400

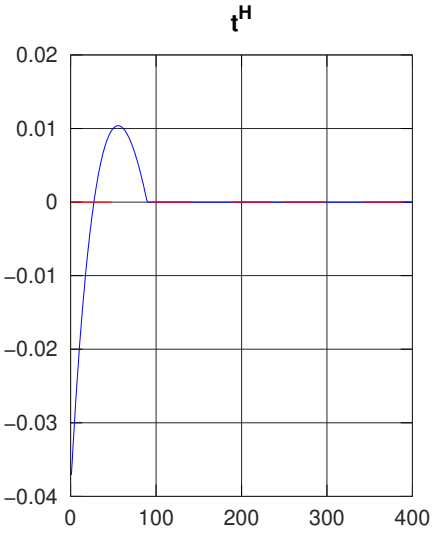

v^F
0
−2
−4
−6
−8
0
100
200
300
400

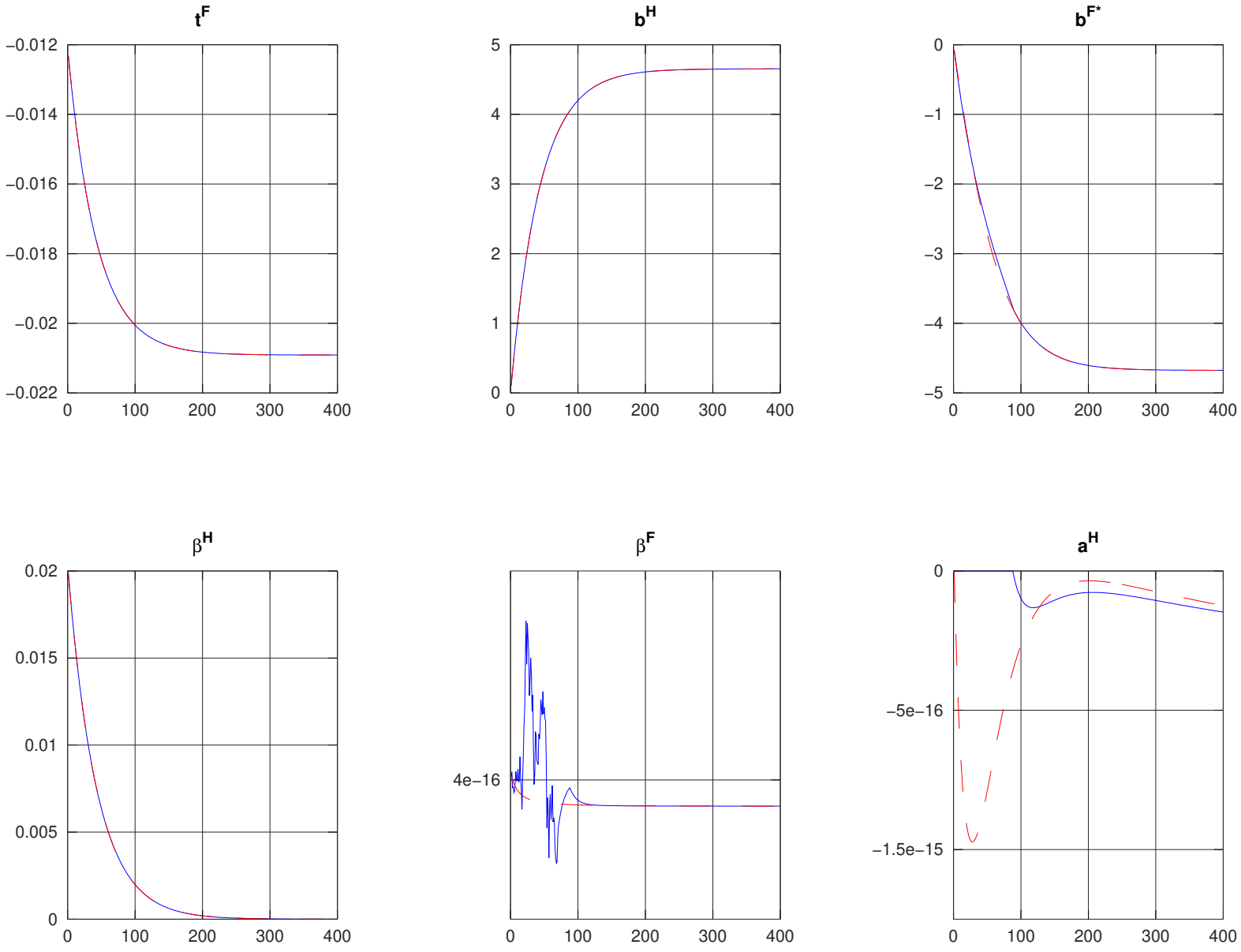

w^H
35
30
25
20
15
10
5
0
0
100
200
300
400

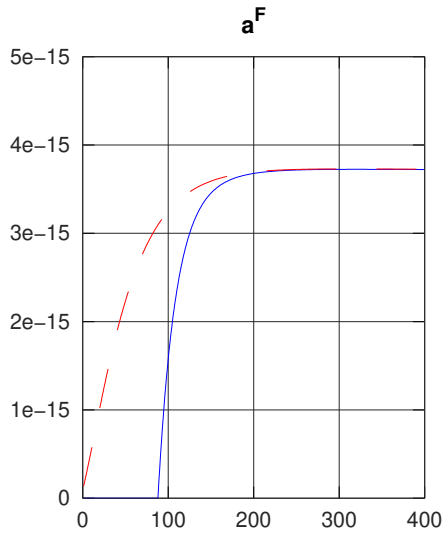

w^F
−0.5
−1
−1.5
−2
−2.5
−3
−3.5
−4
0
100
200
300
400

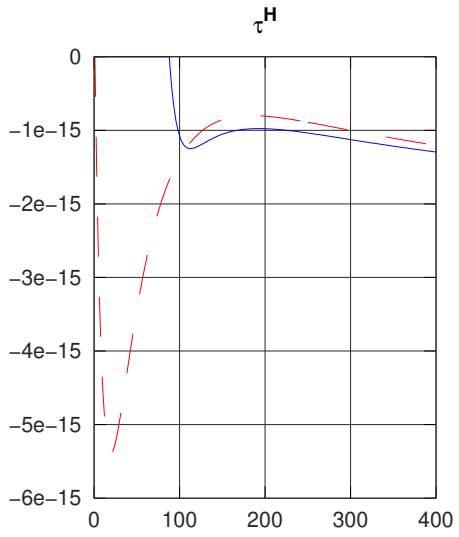

p_bar^F
0.8
0.6
0.4
0.2
0
−0.2
0
100
200
300
400

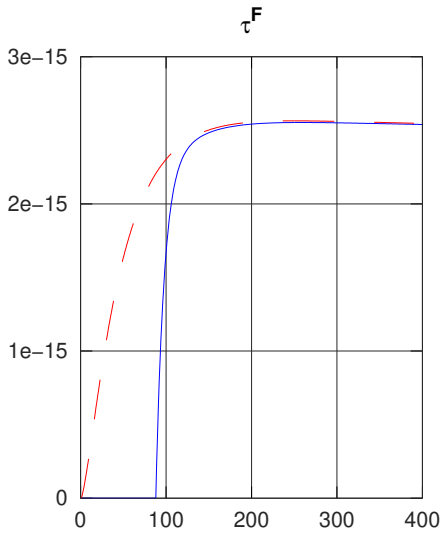

$p_bar^{H^*}$
0.1
0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
−0.7
0
100
200
300
400

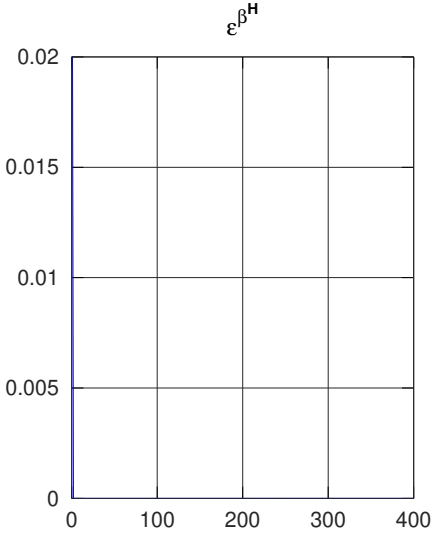

t^H
0.02
0.01
0
−0.01
−0.02
−0.03
−0.04
0
100
200
300
400

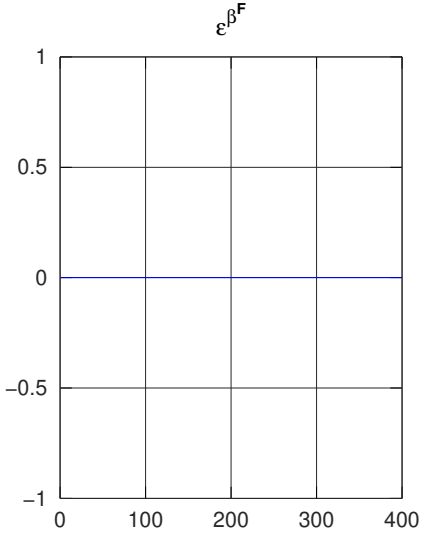

t^F
−0.012
−0.014
−0.016
−0.018
−0.02
−0.022
0
100
200
300
400
b^H
5
4
3
2
1
0
0
100
200
300
400
b^F*
0
−1
−2
−3
−4
−5
0
100
200
300
400
β^H
0.02
0.015
0.01
0.005
0
0
100
200
300
400
β^F
4e−16
0
100
200
300
400
a^H
0
−5e−16
−1.5e−15
0
100
200
300
400

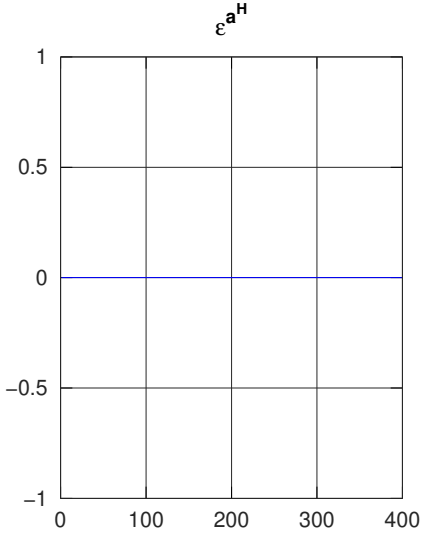

a^F
5e−15
4e−15
3e−15
2e−15
1e−15
0
0
100
200
300
400

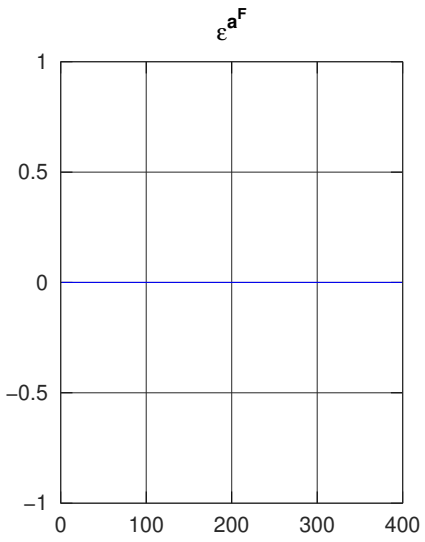

τ^H
0
−1e−15
−2e−15
−3e−15
−4e−15
−5e−15
−6e−15
0
100
200
300
400

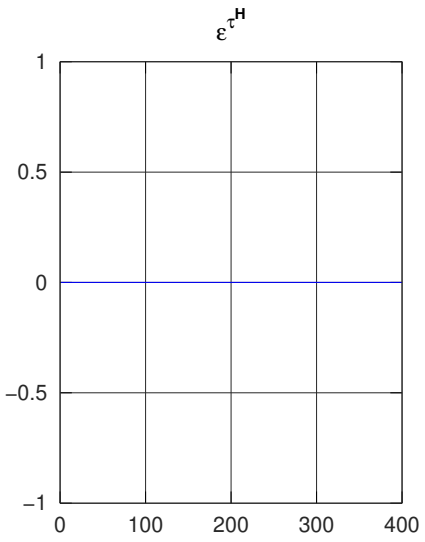

τ^F
3e−15
2e−15
1e−15
0
0
100
200
300
400

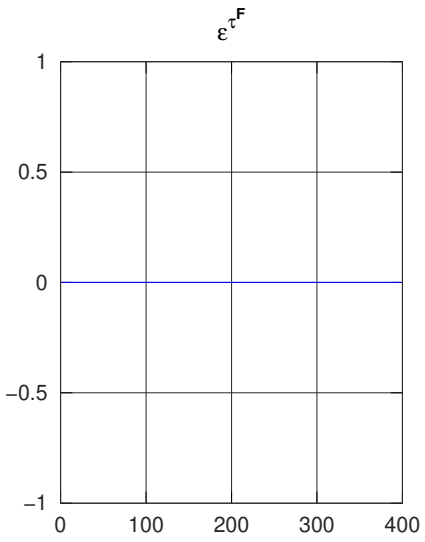

ε^{β^H}
0.02
0.015
0.01
0.005
0
0
100
200
300
400

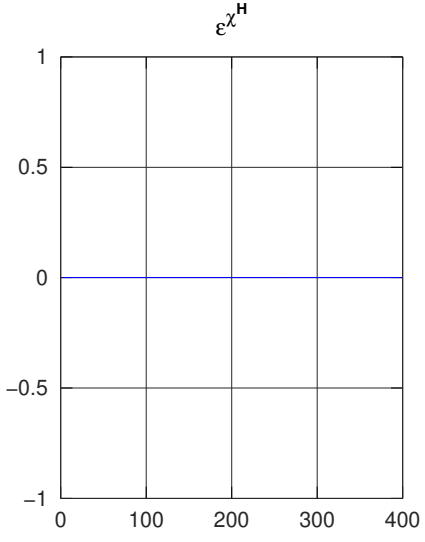

ε^{β^F}
1
0.5
0
−0.5
−1
0
100
200
300
400

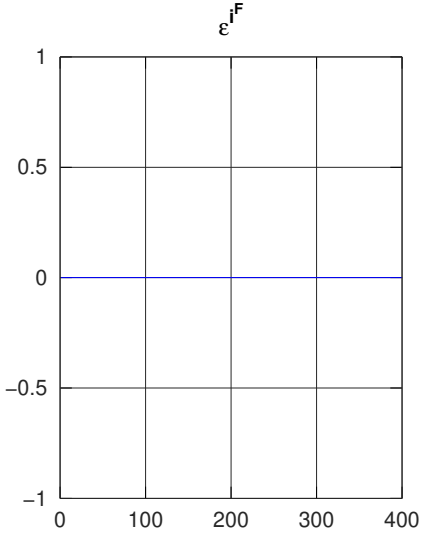

ε^{a^H}
1
0.5
0
−0.5
−1
0
100
200
300
400

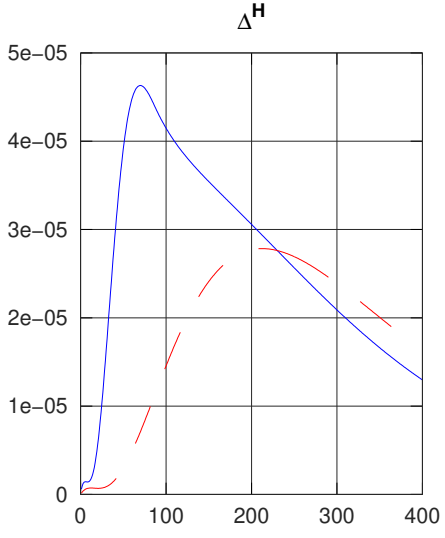

ε^{a^F}
1
0.5
0
−0.5
−1
0
100
200
300
400

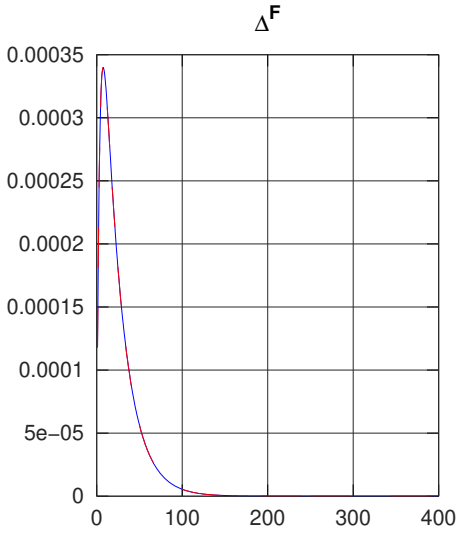

ε^{τ^H}
1
0.5
0
−0.5
−1
0
100
200
300
400


ε^{τ^F}
1
0.5
0
−0.5
−1
0
100
200
300
400


ε^{χ^H}
1
0.5
0
−0.5
−1
0
100
200
300
400


ε^{i^F}
1
0.5
0
−0.5
−1
0
100
200
300
400


Δ^H
5e−05
4e−05
3e−05
2e−05
1e−05
0
0
100
200
300
400


Δ^F
0.00035
0.0003
0.00025
0.0002
0.00015
0.0001
5e−05
0
0
100
200
300
400